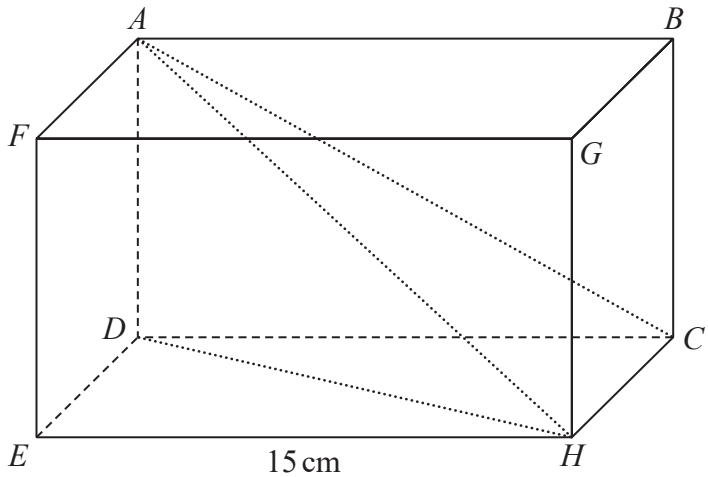
12 *ABCDEFGH* is a cuboid.



Angle *EDH* = 64°
Angle *ACD* = 28°
EH = 15 cm

Work out the size of angle *AHD*.
Give your answer correct to 1 decimal place.

In triangle EDH: sin64° = 15 ÷ DH

DH = 15 ÷ sin64° = 16.68902911...

1 mark
Find DH
using sine.

In triangle ACD: tan28° = AD ÷ 15

AD = 15tan28° = 7.975641475...

1 mark
Find AD
using tangent.

In triangle AHD: tan(angle AHD) = AD ÷ DH

Angle AHD = $\tan^{-1}$(7.975641475... ÷ 16.68902911...)

1 mark
Use AD and DH
in triangle AHD.

= 25.5430082...° = 25.5° (1 decimal place)

1 mark
25.5°.

°

(Total for Question 12 is 4 marks)